SUBSEQUENTIAL INFINITE-VOLUME LIMITS FOR LATTICE GAUGE FIELDS

Abstract. At fixed lattice spacing, compactness of the link configuration space gives subsequential infinite-volume limits and convergence of all polynomial moments of continuous local bounded observables. This argument applies to compact lattice Yang–Mills theory in any dimension and to the link marginal of all-flux chamber QED3. We separate the additional tightness needed to lift the QED limit to real curvature, the moment bounds needed for its correlations, and the estimates uniform in spacing used for a continuum limit. For chamber QED3, a sharp Gaussian estimate supplies these additional bounds.

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1. Introduction

Fix $d \geq 2$, a spacing $a > 0$, and a compact metrizable group G. Let $E^+ = \{(x, i) : x \in \mathbb{Z}^d,\ 1 \leq i \leq d\}$ be the positively oriented nearest-neighbor links and put $\Omega_G = G^{E^+}$. The reverse link carries the inverse group element. A continuous cylinder observable is a continuous function of finitely many links. For $L \geq 5$, extend a probability μ_L on the links of $\mathbb{T}_L = (\mathbb{Z}/L\mathbb{Z})^d$ periodically to a probability $\bar{\mu}_L$ on Ω_G. No density or locality assumption on μ_L is imposed.

Theorem A (Bounded observables)**.** Every sequence $L_n \to \infty$ has a subsequence for which $\bar{\mu}_{L_n} \Rightarrow \mu$ on Ω_G. Along this same subsequence the following conclusions hold.

(i) Expectations of every continuous cylinder observable converge.

(ii) Fix finitely many continuous cylinder maps $\Phi_\alpha : \Omega_G \to \mathbb{R}^k$, and define $B_\alpha(x; U) = \Phi_\alpha(\tau_x U)$, where $(\tau_x U)(y, i) = U(y + x, i)$. The laws of B converge in the coordinate product topology, as do all local polynomial moments of its real components. The maps and their parameters are fixed as $L \to \infty$.

(iii) Let $\mathcal{J}_a B$ equal $B_\alpha(x)$ in component α on $ax + [0, a)^d$. These fields are tempered distributions. For every finite family of Schwartz tests, their pairings converge jointly in law and in every polynomial moment.

(iv) Any continuous transformation of Ω_G preserving all the $\bar{\mu}_L$ preserves μ. If each μ_L is invariant under all torus gauge transformations, μ is invariant under every lattice gauge transformation $U(x, y) \mapsto h(x)U(x, y)h(y)^{-1}$. ■□

The theorem applies to the Wilson laws for $SU(N)$ in dimensions three and four and to the $U(1)$ link marginals of the QED3 laws defined in section 3. For Yang–Mills we use the clover tensor; for QED we use $q^{-1}\operatorname{Im} U_p$ on a plaquette. Both are continuous bounded local observables at fixed parameters. Compactness of the infinite link product is the common argument; the same compactness setup appears in [SZZ23, Section 1.2]. The clover observable follows [Lüs10, Section 3.2].

The QED marginal retains the original photon and fermion weights. Its plaquette holonomy records the real curvature modulo $2\pi/q$; the sine observable retains only its imaginary part. To obtain a law of real curvature, we subsequently extract a joint limit of links and real fields using coordinate tightness. Higher moments require uniform integrability. Sections 4 and 5 give these two steps and the stronger QED Gaussian estimate that supplies both. Section 6 then isolates the estimates whose constants must remain bounded as the spacing tends to zero. The finite QED normalization and Gaussian proofs are in appendices B to D. All volume limits in this note are subsequential; no uniqueness is asserted.

2. Compactness and bounded moments

Choose a complete metric $\rho \le 1$ on G and enumerate $E^+ = \{e_1, e_2, \dots\}$. The metric

$$D(U,V) = \sum_{m\ge 1} 2^{-m}\rho(U_{e_m}, V_{e_m}) \tag{1}$$

induces the product topology on Ω_G. Compactness of G gives a coordinatewise convergent subsequence of every sequence of configurations. The summable tail in (1) then gives convergence in D. Thus Ω_G is compact metrizable. Prokhorov's theorem gives a weakly convergent subsequence of any sequence of its probability laws; see [Sor24, Theorem 1.1]. This proves the extraction in theorem A.

A continuous cylinder observable is bounded on Ω_G, so its expectation converges by weak convergence. Each map Φ_α in the theorem has a finite bound, and translation does not change its range. Hence, for some $M < \infty$,

$$|B_\alpha(x;U)| \le M \qquad (x,\alpha,U). \tag{2}$$

The map from U to the full array B is continuous into the coordinate product. Every polynomial in finitely many components of B is therefore bounded and continuous. The mapping theorem and weak convergence prove the assertions about its law and moments. No independence of the coordinates is needed.

For $Q_x^a = ax + [0,a)^d$, define

$$(\mathcal{J}_a B)_\alpha(y) = B_\alpha(x), \qquad y \in Q_x^a. \tag{3}$$

Pairing with a Schwartz test $\varphi = (\varphi_\alpha)$ gives

$$\langle \mathcal{J}_a B, \varphi\rangle = \sum_{x,\alpha} \left\langle B_\alpha(x), \int_{Q_x^a} \varphi_\alpha(y)\, dy \right\rangle.$$

The series is bounded absolutely by $M\sum_\alpha \|\varphi_\alpha\|_{L^1}$, uniformly in the configuration. It defines a tempered distribution. Its finite truncations are continuous cylinder functions; the remainder is bounded uniformly by the corresponding L^1 tail of the test. Thus the full pairing is continuous and bounded on Ω_G. A finite vector of such pairings is also continuous and bounded, so its law and every polynomial moment converge along the selected subsequence.

If a continuous map T preserves every $\bar\mu_L$, pass $\int f\circ T\, d\bar\mu_L = \int f\, d\bar\mu_L$ to the limit for $f \in C(\Omega_G)$. This proves the first symmetry assertion. For a lattice gauge transformation h, take a continuous cylinder f and choose L so large that all endpoints of its links embed in $\mathbb{T}_L$. A torus gauge transformation can then be prescribed to equal h at these endpoints. Finite gauge invariance gives

$$\int f(T_h U)\, d\bar\mu_L(U) = \int f(U)\, d\bar\mu_L(U).$$

Both integrands are continuous cylinders. The identity passes to the limit. Continuous cylinders determine measures on Ω_G: they are uniformly dense in $C(\Omega_G)$, by uniform continuity and the summable tail of D. Hence the limiting law is invariant under every h. This completes theorem A.

3. Yang–Mills and QED link laws

For links written by their endpoints, let $U(x,y)$ denote the variable from x to y and set $U(y,x) = U(x,y)^{-1}$. The plaquette holonomy based at x is

$$U_{ij}(x) = U(x,x+e_i)U(x+e_i,x+e_i+e_j)U(x+e_j,x+e_i+e_j)^{-1}U(x,x+e_j)^{-1}. \tag{4}$$

3.1. **Compact Yang–Mills theory.** Let H be normalized Haar measure on G and let $\phi : G \to \mathbb{R}$ be continuous, conjugation invariant, and satisfy $\phi(g^{-1}) = \phi(g)$. At fixed $\beta \in \mathbb{R}$, set

$$d\mu_L = Z_L^{-1} \exp\left(-\beta \sum_{p \in P_L^+} \phi(U_p)\right) \prod_{e \in E_L^+} dH(U_e). \tag{5}$$

Here P_L^+ contains each unoriented elementary square once. Boundedness of ϕ gives $0 < Z_L < \infty$. Haar invariance and the assumptions on ϕ give translation and gauge invariance. Theorem A therefore applies. For the Wilson action, $G = SU(N)$ and $\phi(g) = 1 - N^{-1} \operatorname{Re}\operatorname{tr} g$. There is no restriction to strong coupling for this extraction.

Identify $\mathfrak{su}(N)$ with traceless anti-Hermitian matrices. For signed coordinate vectors $v \perp w$, put

$$U_v(x) = U(x,x+v), \qquad P(v,w;x) = U_v(x)U_w(x+v)U_v(x+w)^{-1}U_w(x)^{-1}.$$

The clover average and its field tensor are

$$\begin{aligned} \mathcal{C}_{ij}(x) &= \tfrac{1}{4}\big(P(e_i,e_j;x) + P(e_j,-e_i;x) + P(-e_i,-e_j;x) + P(-e_j,e_i;x)\big), \\ F_{ij}^{(a)}(x) &= a^{-2}\Pi_{\mathfrak{su}(N)}\mathcal{C}_{ij}(x), \qquad \Pi_{\mathfrak{su}(N)}X = \tfrac{1}{2}(X - X^*) - \tfrac{1}{2N}\operatorname{tr}(X - X^*)I. \end{aligned} \tag{6}$$

All four loops start at x with the same orientation. Reversing i,j takes the adjoint, so $F_{ji}^{(a)} = -F_{ij}^{(a)}$. This is the clover discretization of [Lüs10, Section 3.2 and Figure 1]. Each component is a continuous function of finitely many links, and

$$\|F_{ij}^{(a)}(x)\|_{\mathrm{op}} \le 2a^{-2}. \tag{7}$$

Based plaquette products transform by conjugation at x, whence

$$F_{ij}^{(a)}(T_h U;x) = h(x)F_{ij}^{(a)}(U;x)h(x)^{-1}. \tag{8}$$

Taking real matrix components, theorem A proves convergence of the clover laws and all their local polynomial and Schwartz-pairing moments. This applies in particular to YM3 and YM4. Finite Wilson loops and curvature insertions joined by finite parallel transports are also continuous local observables and pass to the same limit.

Multiplying (5) by any nonnegative Borel matter weight with finite, nonzero integral leaves these compactness and moment conclusions valid after normalization. This includes nonnegative fermion determinants. Symmetries require the corresponding invariance of the weight. For the pure plaquette action, the limiting law also satisfies the local Gibbs equations of proposition A.1. That additional conclusion uses the specified action.

3.2. **The bounded observables of chamber QED3.** Fix $q > 0$, $r = 2\pi/q$, and $d = 3$. On a cubic torus, write $A \in [0, r)^{E_L^+}$ and let z run over all closed integer two-cochains. The real curvature and compact links are

$$F = dA + rz, \qquad U_e = e^{iqA_e}, \qquad U_p = e^{iqF_p}. \tag{9}$$

The normalized finite law is

$$d\widehat{\mu}_L(A, z) = Z_L^{-1} W_L(A) e^{-\|dA+rz\|^2/2}\, dA, \qquad dz = 0, \tag{10}$$

where integration includes counting measure in z. Here W_L is nonnegative, r-periodic in every link, and $0 < I_L := \int_{[0,r)^{E_L^+}} W_L(A) dA < \infty$. These conditions give positive finite normalization. Indeed, retain $z = 0$ for positivity, and for finiteness drop $dz = 0$ and use

$$\sup_y \sum_{k\in\mathbb{Z}} e^{-(y+rk)^2/2} < \infty$$

on each plaquette before integrating W_L.

For chamber QED3, $W_L(A) = \det \mathcal{D}_A$, with $0 < \rho < 1$, $w_0, w_1, w_2, w_3 > 0$, $M \in \mathbb{R}$, and antiperiodic spinors in all three coordinate directions. The operator and its normalization for every real M are specified and proved in appendix B and proposition B.1. The case $W_L = 1$ gives the corresponding pure photon law.

Push (10) forward by $A \mapsto U$ to obtain μ_L on $U(1)^{E_L^+}$. This is the compact link marginal of the same all-flux law. In phase coordinates its density is proportional to

$$W_L(A) \sum_{dz=0} e^{-\|dA+rz\|^2/2}.$$

It need not have a Wilson plaquette action. The proof of theorem A requires no such representation.

For each positively oriented plaquette set

$$B_p = q^{-1} \operatorname{Im} U_p = \frac{\sin(qF_p)}{q}, \qquad |B_p| \le q^{-1}. \tag{11}$$

This is a continuous local observable of the compact links. The theorem gives a link limit, its B-field law, and convergence of all local polynomial and Schwartz-pairing moments of B, along the same subsequence. The chamber laws have the translation and gauge symmetries verified in appendix B, which therefore pass to the limit. The bounded-observable proof ends here.

The real F_p in (9) includes an integer lift absent from U_p. The bounded field B generally does not satisfy the linear equation $dB = 0$. Its limit therefore specifies neither a unique real lift nor the moments of that lift. The next section gives the additional estimates for a joint limit retaining F.

4. Recovering a real-curvature law

Let $\mathcal{P} = \{(x; i, j) : x \in \mathbb{Z}^d,\ i < j\}$ and fix $q > 0$. Suppose that $\widehat{\mu}_L$ are joint laws of compact $U(1)$ links and real plaquette fields, extended periodically, with

$$U_p = e^{iqF_p}, \qquad dF = 0. \tag{12}$$

Here dF is the oriented sum over cube faces. Let a subsequence of the link marginals converge to μ as in theorem A.

Proposition 4.1. Assume that the laws of F_p are uniformly tight in L for each fixed $p \in \mathcal{P}$. Then a further subsequence of the joint laws converges on $\Omega_{U(1)} \times \mathbb{R}^{\mathcal{P}}$ to a probability $\widehat{\mu}$ with link marginal μ, supported on (12). If, in addition, for every integer $m \ge 1$,

$$\sup_{L,p} \mathbb{E}_{\widehat{\mu}_L} |F_p|^m \le C_m < \infty, \tag{13}$$

then all local polynomial moments of F converge along that subsequence, also with any bounded continuous link cylinder factor. At each fixed spacing, the step fields $\mathcal{J}_a F$ are tempered, and their Schwartz pairings converge jointly in law and in every polynomial moment. ■

Proof. Enumerate $\mathcal{P} = \{p_1, p_2, \ldots\}$. For $\varepsilon > 0$, coordinate tightness provides R_j such that $\sup_L \widehat{\mu}_L(|F_{p_j}| > R_j) \le \varepsilon 2^{-j}$. The set

$$\Omega_{U(1)} \times \prod_{j \ge 1} [-R_j, R_j]$$

is compact and has probability at least $1 - \varepsilon$ under every joint law. Prokhorov's theorem gives a joint weak subsequence. Projection identifies its link marginal with μ. Both relations in (12) are closed coordinate conditions, so they pass to the limit. This proves existence of a lift; it does not assert uniqueness of the lift of μ.

For a polynomial P of degree k supported on a finite set S,

$$|P(F)|^2 \le C_P \left(1 + \sum_{p \in S} |F_p|^{2k}\right).$$

Thus (13) gives uniform integrability of $P(F)$, also after multiplication by a bounded link cylinder. The corresponding bounds in the limit follow by bounded truncation and monotone convergence. Clip P continuously to a bounded function, pass expectations by weak convergence, and remove the clipping using uniform integrability. This proves the local moment assertion. For that assertion alone, the bounds may depend on p.

The spatial uniformity in (13) also gives, for $s > d/2$,

$$\sup_L \mathbb{E} \sum_{x, i<j} (1 + |x|^2)^{-s} |F_{ij}(x)|^2 \le C_2 \binom{d}{2} \sum_x (1 + |x|^2)^{-s} < \infty.$$

The same estimate holds in the limit. Tonelli and Cauchy–Schwarz show that the step fields are almost surely tempered. For the summable coefficients $b_{x,ij} = \int_{Q_x^a} \varphi_{ij}(y)dy$ of a Schwartz pairing, Minkowski gives

$$\sup_L \left\| \sum_{p \notin S} b_p F_p \right\|_{L^m} \le C_m^{1/m} \sum_{p \notin S} |b_p| \longrightarrow 0 \tag{14}$$

as finite sets S exhaust $\mathcal{P}$; the same holds for the limit. Finite sums converge in law by joint weak convergence. The uniform L^1 tails then give joint convergence for the full pairings. Their higher L^m bounds and Hölder's inequality give uniform integrability of every polynomial in a finite family of pairings, completing the proof. □

A uniform positive moment of each coordinate suffices for the first hypothesis. In particular, a second-moment estimate can establish a real lift without establishing convergence of its quadratic moments. The latter requires uniform integrability; the all-order bounds above provide it for every polynomial degree.

The same moment argument applies to finite-dimensional real vector fields, without (12). Thus it also covers the matrix components of the YM clover tensor. In that case the deterministic bound (7) already supplies every moment bound. Closedness in (12) is a condition on the real Abelian lift, not on the non-Abelian clover tensor.

5. Gaussian estimates for the real field

The preceding lifting argument requires only coordinate tightness. A Gaussian estimate supplies the additional moments and a bound on all linear tests that will also be useful when the spacing

varies. A real two-cochain is an array $F \in \mathbb{R}^{\mathcal{P}}$. The lattice coboundary dF on a cube is the oriented sum over its faces. Consequently,

$$\Omega_{\mathrm{cl}} = \{F \in \mathbb{R}^{\mathcal{P}} : dF = 0\}$$

is closed in the product topology. The subscript denotes lattice closedness. It does not assert that the step interpolation (3) is closed under the continuum exterior derivative.

For each L, let λ_L be a probability law on real two-cochains on $\mathbb{T}_L$, supported on $dF = 0$, and denote its periodic extension law by $\bar{\lambda}_L$. Both finite and infinite lattice pairings below use counting measure on positively oriented plaquettes.

Theorem B. Suppose that, for a fixed $\kappa > 0$, every $L \geq 4$ and every real torus two-cochain b satisfy

$$\mathbb{E}_{\lambda_L} \exp\langle b, F\rangle \leq \exp(\kappa \|b\|^2/2). \tag{15}$$

Then the laws $\bar{\lambda}_L$ are tight on Ω_{cl}. Every sequence $L_n \to \infty$ has a subsequence converging weakly to a probability λ on this space. Every such limit satisfies

$$\mathbb{E}_{\lambda} e^{\langle b,F\rangle} \leq e^{\kappa \|b\|_{\ell^2}^2/2} \qquad (b \text{ real and finitely supported}). \tag{16}$$

Along the same subsequence all local polynomial moments converge. For every $s > d/2$ the limiting field is almost surely in the weighted Hilbert space

$$\mathcal{H}_{-s} = \left\{ F : \|F\|_{-s}^2 := \sum_{x \in \mathbb{Z}^d} \sum_{i<j} (1 + |x|^2)^{-s} |F_{ij}(x)|^2 < \infty \right\}. \tag{17}$$

For each $t > d/2$ the same subsequence converges weakly as laws on $\mathcal{H}_{-t}$. At every fixed spacing $a > 0$, the interpolated fields $\mathcal{J}_a F$ are tempered distributions; all their finite families of Schwartz pairings converge in law and in joint moments. Any translation or continuous signed-permutation symmetry shared by the finite-volume extended laws is inherited by the limit. ■□

5.1. **Tails and compact sets.** Apply (15) to a test supported on one torus plaquette. For $u > 0$, Markov's inequality with $t = u/\kappa$ gives

$$\bar{\lambda}_L(|F_p| > u) \leq 2e^{-u^2/(2\kappa)} \qquad (p \in \mathcal{P}). \tag{18}$$

There is no requirement here that several lattice plaquettes embed distinctly on the torus: only one coordinate is being tested. The estimate is uniform in both L and the location of p.

The product compact-set argument in proposition 4.1, with the link factor omitted, now proves tightness and subsequential weak convergence. The closed coordinate condition $dF = 0$ passes to the limit, so the limiting law is supported on Ω_{cl}.

5.2. **Passage of exponential bounds and moments.** Fix a finitely supported real lattice cochain b. For all sufficiently large L, its support embeds injectively in the torus. Folding b to that torus then preserves its counting norm and its pairing with the periodic field. Thus

$$\mathbb{E}_{\bar{\lambda}_L} e^{\langle b,F\rangle} \leq e^{\kappa \|b\|_{\ell^2}^2/2}$$

for all such L. For $R > 0$, the function $\min\{R, e^{\langle b,F\rangle}\}$ is a bounded continuous cylinder. Weak convergence and then monotone convergence as $R \to \infty$ prove (16).

Here and below the sharp norm is used only after this support embedding. For arbitrary L, the folded test is $b_L(p) = \sum_{\tilde{p} \mapsto p} b(\tilde{p})$, and the finite estimate contains $\|b_L\|$, which can exceed $\|b\|_{\ell^2}$.

If a real random variable X satisfies $\mathbb{E}e^{tX} \leq e^{t^2 v^2/2}$ for all real t, then

$$\mathbb{E}e^{t|X|} \leq 2e^{t^2 v^2/2} \quad (t \geq 0). \qquad \mathbb{P}(|X| > u) \leq 2e^{-u^2/(2v^2)}.$$

The exponential estimate justifies differentiating the moment generating function at zero. Comparing its linear and quadratic terms with the upper bound for both signs of t gives $\mathbb{E}X = 0$ and $\mathbb{E}X^2 \le v^2$. For every $p > 0$, integration of the tail gives

$$\mathbb{E}|X|^p \le c_p v^p, \qquad c_p = 2\,2^{p/2}\Gamma(1 + p/2). \tag{19}$$

Apply this with $X = \langle b, F\rangle$ and $v^2 = \kappa\|b\|_{\ell^2}^2$, either under the limiting law or on a sufficiently large torus. In particular the limiting linear field is centered.

The coordinate bounds imply (13). The truncation proof in proposition 4.1, which does not use the link variables, proves convergence of all local polynomial moments. More generally,

$$\mathbb{E}_\lambda \prod_{\alpha=1}^{m} |\langle b_\alpha, F\rangle| \le c_m \kappa^{m/2} \prod_{\alpha=1}^{m} \|b_\alpha\|_{\ell^2}.$$

This also extends the limiting linear field continuously from finite cochains to $\ell^2(\mathcal{P})$ with values in every L^p, $p \ge 1$, by completion.

5.3. **Weighted compactness and distributional realization.** The second-moment estimate at a single coordinate gives, for $s > d/2$,

$$\sup_L \mathbb{E}_{\bar\lambda_L} \|F\|_{-s}^2 \le \kappa \binom{d}{2} \sum_{x\in\mathbb{Z}^d} (1 + |x|^2)^{-s} < \infty. \tag{20}$$

Tonelli's theorem proves that the approximating fields lie in $\mathcal{H}_{-s}$ almost surely. Applying the limiting one-coordinate bound proves the same assertion for λ. The norm is a measurable countable sum of coordinate functions; these statements therefore define probabilities on the separable weighted Hilbert spaces.

For $t > s$, the inclusion $\mathcal{H}_{-s} \hookrightarrow \mathcal{H}_{-t}$ is compact. Indeed, on a ball $\|F\|_{-s} \le R_0$, the squared $\mathcal{H}_{-t}$ norm outside $|x| \le R$ is at most $R_0^2(1 + R^2)^{-(t-s)}$. Inside that finite set, a diagonal selection gives convergence. The uniform tail estimate then gives convergence in $\mathcal{H}_{-t}$. Fatou's lemma keeps the limit in the original ball, so its image is compact. Markov's inequality and (20) prove tightness in $\mathcal{H}_{-t}$ whenever $t > s > d/2$.

Fix $t > d/2$ and choose s strictly between $d/2$ and t. Any subsequence of the already selected product-convergent sequence has a further subsequence converging in law on $\mathcal{H}_{-t}$. Coordinate evaluations are continuous on that Hilbert space, so every such limit has the same coordinate law λ. Coordinates generate its Borel sigma algebra: after multiplication by the square roots of the weights, they are the coordinates of ℓ^2. The Hilbert-space law is therefore unique. Tightness then proves convergence of the entire selected subsequence on $\mathcal{H}_{-t}$. No further extraction is needed.

For a Schwartz test φ, put $B_{x,ij} = \int_{Q_x^a} \varphi_{ij}(y)\,dy$. For every t the rapid decay of φ implies

$$\sum_{x,i<j} (1 + |x|^2)^t |B_{x,ij}|^2 < \infty.$$

Cauchy–Schwarz therefore gives

$$|\langle \mathcal{J}_a F, \varphi\rangle| \le \|F\|_{-t} \left(\sum_{x,i<j} (1 + |x|^2)^t |B_{x,ij}|^2 \right)^{1/2}. \tag{21}$$

The second factor is bounded by a Schwartz seminorm, with a constant depending on the fixed a, t, d. Thus $\mathcal{J}_a F$ is tempered and its test pairings are continuous on $\mathcal{H}_{-t}$. Their finite-dimensional laws converge by the mapping theorem.

The uniform one-coordinate bounds and (14) give convergence of all joint moments of these pairings, as in proposition 4.1.

Finally, any shared lattice symmetry in the theorem is a continuous coordinate permutation, possibly followed by signs. For bounded continuous f, pass the identity $\int f \circ T \, d\bar{\lambda}_L = \int f \, d\bar{\lambda}_L$ to the limit. This proves the symmetry assertion and completes theorem B.

5.4. **The additional QED estimate.** For the chamber law of appendix B, the finite-volume statement is

$$\mathbb{E}_{\mathbf{n}} e^{\langle b,F\rangle} \le e^{\|b\|^2/2} \quad (b \in C^2(C_{\mathbf{n}};\mathbb{R}),\ n_i \ge 4). \tag{22}$$

It holds for every parameter choice (35); proposition D.3 gives the complete statement and proof. This estimate is used to recover the real-field moments and to control the physical rescaling, rather than to extract the compact link law. Its proof uses the particular fermion operator and the quadratic Maxwell action.

Completing the square reduces the estimate to $Z_{\mathbf{n}}(b) \le Z_{\mathbf{n}}(0)$. The finite fermion factorization in appendix C yields an exact slice trace. Electric shifts enter through unitaries, while magnetic shifts enter through multiplication operators. Schatten's inequality removes the unitaries and repeats a single magnetic slice. In three dimensions only one magnetic component remains; a second choice of time direction makes it electric and removes it. The all-flux unfolding and Hilbert–Schmidt estimates in appendix D justify the trace and its comparisons.

Theorem C. Fix the parameters (35), with the chamber operator, all-flux photon measure, and antiperiodic spinors specified in appendix B. The periodic curvature laws on $C_{(L,L,L)}$, $L \ge 4$, have subsequential infinite-volume limits along every sequence $L \to \infty$. Every limit is a probability on closed real lattice two-cochains and satisfies

$$\mathbb{E} e^{\langle b,F\rangle} \le e^{\|b\|_{\ell^2}^2/2} \quad (b \text{ finitely supported and real}).$$

It is invariant under translations, signed coordinate permutations with their cochain signs, and $F \mapsto -F$. All local polynomial moments converge along the selected subsequence. The weighted-space, tempered-field and Schwartz-moment conclusions of theorem B hold with $d = 3$ and $\kappa = 1$. ■□

Proof. Propositions B.1 and D.3 give normalized finite laws and (15) with $\kappa = 1$. Apply theorem B. The finite symmetries are verified in appendix B. If a compact-link subsequence was selected first, proposition 4.1 gives a further joint subsequence, whose real marginal has these conclusions and whose link marginal is the previously selected limit. Thus the compact and real laws can be obtained together without asserting that the compact law determines its real lift. □

6. Estimates when the spacing tends to zero

The bounds in theorem A may depend on the fixed spacing and bare parameters. For YM clover curvature, (7) grows like a^{-2}. In the QED3 scaling, $q_a = e\sqrt{a}$ and the physical real field is $a^{-3/2}F$ on a cell of side a. The corresponding sine observable has the deterministic bound

$$\left| a^{-3/2} \frac{\sin(q_a F_p)}{q_a} \right| \le \frac{1}{ea^2}. \tag{23}$$

These fixed-spacing bounds alone give no uniform control of continuum field-strength moments.

6.1. **Uniform estimates on smeared real fields.** Take the infinite-volume real QED law first, at each spacing a, with the parameters of theorem C, allowing these parameters to depend on a. The Gaussian constant remains one. For a real Schwartz two-form f on $\mathbb{R}^3$, put

$$b_a(x;i,j) = a^{-3/2} \int_{ax+[0,a)^3} f_{ij}(y)\, dy, \qquad X_a(f) = \langle b_a, F\rangle. \tag{24}$$

The pairing is the real field's physical step interpolation. Cauchy–Schwarz on each cell gives

$$\|b_a\|_{\ell^2}^2 \le \|f\|_{L^2}^2. \tag{25}$$

Finite truncations of b_a converge in ℓ^2. The Gaussian moment bounds give convergence of their pairings in every L^p; Fatou passes the exponential inequality. Consequently

$$\mathbb{E}e^{tX_a(f)} \le e^{t^2\|f\|_2^2/2}, \tag{26}$$

$$\mathbb{E}|X_a(f)|^p \le c_p\|f\|_2^p \quad (p>0), \tag{27}$$

$$\left|\mathbb{E}e^{iX_a(f)} - \mathbb{E}e^{iX_a(g)}\right| \le \|f-g\|_2. \tag{28}$$

Here c_p is the constant in (19). All three estimates are uniform in a and in the chosen volume limit. For the last inequality use $|e^{iu}-e^{iv}| \le |u-v|$ and $\mathbb{E}|X_a(f-g)|^2 \le \|f-g\|_2^2$. The order of limits matters in this deduction: the infinite lattice test has no periodic folding, so (25) is the relevant norm estimate.

The second-moment bound supplies distributional tightness through weighted Hermite coefficients. To spell out the estimate, let h_ν, $\nu \in \mathbb{N}_0^3$, be the orthonormal Hermite basis of $L^2(\mathbb{R}^3)$, and use one copy for each two-form component. Write $|\nu| = \nu_1+\nu_2+\nu_3$. For $s > 3/2$,

$$\sup_a \mathbb{E}\sum_{i<j}\sum_{\nu\in\mathbb{N}_0^3}(1+|\nu|)^{-2s}|X_a(h_\nu\, dy_i \wedge dy_j)|^2 \le 3\sum_{\nu\in\mathbb{N}_0^3}(1+|\nu|)^{-2s} < \infty. \tag{29}$$

The associated weighted coefficient space embeds compactly into the space with exponent $t > s$: a bounded ball has squared tail at most a constant times $(1+R)^{-2(t-s)}$ outside $|\nu| \le R$. Markov's inequality therefore gives tightness in the latter space. Polynomially growing Hermite coefficients define tempered distributions, since Schwartz tests have rapidly decreasing Hermite coefficients; see [RT07, Sections 2 and 5] for the Hermite Sobolev description of these spaces. Thus the real physical fields have subsequential distributional limits. Pairings with Schwartz tests are continuous on these weighted spaces, and (27) gives uniform integrability of their products. Every selected limit therefore inherits the Gaussian bound and convergence of all joint Schwartz-pairing moments.

This deduction concerns compactness and moments. Identifying the remaining field equations, covariance, positivity, and interaction of a continuum law requires their respective estimates. No continuum Yang–Mills conclusion is obtained from the bounded clover result in section 3.

6.2. **Comparing a bounded observable with real curvature.** Let $Y_a(f)$ be the physically rescaled pairing of a chosen bounded QED observable after the volume limit. To identify its continuum moments with those of X_a, a sufficient comparison is

$$\|Y_a(f) - X_a(f)\|_{L^p} \longrightarrow 0 \quad \text{for every Schwartz } f \text{ and every } p \ge 1, \tag{30}$$

under a joint realization, such as the lifted law of proposition 4.1. Together with (27), this gives convergence of joint laws and moments by a telescoping expansion of products and Hölder's inequality.

Equation (30) is an additional estimate; it is not a conclusion of the compact-link theorem. For the sine observable, the elementary inequality

$$\left|u - \frac{\sin(qu)}{q}\right| \le \frac{q^2|u|^3}{6}$$

and a uniform one-coordinate third moment yield only a bound proportional to $a^{-3/2}q_a^2\|f\|_{L^1}$ after smearing. At $q_a = e\sqrt{a}$ this is of order $a^{-1/2}$, so it does not establish the required comparison. The real-field continuum compactness above uses the real lift and its Gaussian estimate directly; it does not require identifying the sine observable with that continuum field.

Appendix A. Local Gibbs equations for plaquette actions

For the pure plaquette laws (5), the compact limit also retains the finite-volume conditional distributions. This property uses the action and is separate from compactness.

Proposition A.1. Let G be compact metrizable, $\beta \in \mathbb{R}$, and $\phi : G \to \mathbb{R}$ continuous, conjugation invariant, and invariant under inversion. Every periodic weak limit of (5) is translation and gauge invariant. For each finite $\Delta \subset E^+$, its conditional law on Δ, given the exterior links, has density proportional to

$$\exp\left(-\beta \sum_{p:\, \partial p \cap \Delta \neq \varnothing} \phi(U_p)\right) \prod_{e \in \Delta} dH(U_e),$$

where the intersection ignores link orientation. Its bounded local observable and moment conclusions are those of theorem A. ■

The symmetries follow from the finite changes of variables in section 3 and theorem A. We prove the conditional assertion using cylinder tests before taking the limit.

A.1. **Conditional integration in a finite set.** Fix a finite set $\Delta \subset E^+$. Let $\mathcal{P}(\Delta)$ be the finite collection of elementary squares containing at least one unoriented edge from Δ. For an exterior configuration η and an assignment $v \in G^\Delta$, let $v\eta$ denote the configuration equal to v on Δ and to η elsewhere. Define

$$H_\Delta(v \mid \eta) = \beta \sum_{p \in \mathcal{P}(\Delta)} \phi((v\eta)_p), \tag{31}$$

$$Z_\Delta(\eta) = \int_{G^\Delta} e^{-H_\Delta(v|\eta)} \prod_{e \in \Delta} dH(v_e), \tag{32}$$

$$(\gamma_\Delta f)(\eta) = \frac{\int_{G^\Delta} f(v\eta) e^{-H_\Delta(v|\eta)} \prod_{e\in\Delta} dH(v_e)}{Z_\Delta(\eta)}. \tag{33}$$

The denominator is strictly positive. With $b_\Delta = |\beta| \|\phi\|_\infty |\mathcal{P}(\Delta)|$,

$$e^{-b_\Delta} \le Z_\Delta(\eta) \le e^{b_\Delta} \qquad \text{for every } \eta.$$

For a continuous cylinder function f, $\gamma_\Delta f$ is again a continuous cylinder function. Only the variables read by f and by the plaquettes in $\mathcal{P}(\Delta)$ can occur. Continuity of the numerator follows by integration of a continuous function on a finite compact product; the denominator has the same property and is bounded away from zero.

Let g be a continuous cylinder function reading only edges outside Δ. If L is large enough, reduction modulo L is injective on all vertices and edges needed by Δ, f, g, and $\mathcal{P}(\Delta)$. Increase L if necessary so that every torus square touching the image of Δ is the image of exactly one square in $\mathcal{P}(\Delta)$. Finite-dimensional conditioning under (5) gives

$$\int g f \, d\bar{\mu}_L = \int g \, \gamma_\Delta f \, d\bar{\mu}_L. \tag{34}$$

To verify the identity, integrate first in the coordinates corresponding to Δ. The terms in the torus action not touching these coordinates are constant in that integration. Their factors cancel between the conditional numerator and denominator, leaving exactly (33).

This argument uses finite cylinder tests. The full infinite exterior of a periodically extended configuration includes remote copies of every variable in Δ. Conditioning that periodic law on its entire exterior would therefore determine the interior variables. Equation (34) asserts a finite-test identity on a sufficiently large torus, which is the identity that survives the limit.

Suppose $\bar{\mu}_{L_n} \Rightarrow \mu$ along the selected subsequence. Both sides of (34) have bounded continuous integrands. Passing to the limit gives the same identity with μ in place of $\bar{\mu}_L$. For fixed continuous cylinder f, the functional monotone-class theorem extends it from continuous exterior cylinder g to every bounded exterior-measurable g. These cylinders generate the exterior product sigma algebra.

Now fix such a g. The class of bounded Borel f for which the identity holds is closed under bounded pointwise convergence, by dominated convergence on the measure and kernel sides. Continuous

cylinders form a unital algebra generating the Borel sigma algebra of Ω_G. A second monotone-class argument proves the identity for every bounded Borel f. The function $\gamma_\Delta f$ is exterior-measurable. We have proved

$$\mathbb{E}_\mu[f \mid \sigma(U_e : e \notin \Delta)] = \gamma_\Delta f \quad \mu\text{-almost surely.}$$

This is the claimed local Gibbs equation.

A.2. **Other local actions and other exhaustions.** The proof extends to finite sums of translated loop interactions. Take finitely many finite closed lattice paths $\ell_1, \dots, \ell_m$ and continuous real class functions $\phi_1, \dots, \phi_m$. Replace the plaquette action by

$$\sum_{\alpha=1}^{m} \sum_{x \in \mathbb{T}_L} c_\alpha \phi_\alpha(U_{x+\ell_\alpha}), \qquad c_\alpha \in \mathbb{R}.$$

For all sufficiently large L, the paths are defined without local identifications. Every fixed update set meets only finitely many translated interaction terms. Finite normalization, continuity of the conditional kernels, and the local embedding argument are unchanged. Gauge and translation invariance hold because the interactions are closed-loop class functions and their translates are summed with constant coefficients.

There is also an existence result for fixed exterior links. Let finite edge sets Λ_n exhaust E^+, and let η_n be arbitrary exterior configurations. The probabilities $\gamma_{\Lambda_n}(\cdot \mid \eta_n)$ on Ω_G have subsequential limits by compactness. For each fixed finite update Δ and cylinder tests, the conditional identity holds once Λ_n contains Δ and its interaction neighborhood. Every limit is therefore a Gibbs law for the same local interaction. Translation and gauge invariance are not asserted for these boundary-selected limits, and this argument does not identify them with limits selected by periodic boundary conditions.

Appendix B. The finite QED3 curvature law

This appendix defines the chamber operator used in section 3 and proves positive finite normalization for every real mass. The subsequent two appendices supply the additional Gaussian estimate (22).

B.1. **Photon configurations and the chamber operator.** Fix

$$q > 0, \qquad 0 < \rho < 1, \qquad w_0, w_1, w_2, w_3 > 0, \qquad M \in \mathbb{R}, \qquad r_q = 2\pi/q. \tag{35}$$

Let $C_{\mathbf{n}} = \mathbb{Z}^3/(n_1\mathbb{Z} \oplus n_2\mathbb{Z} \oplus n_3\mathbb{Z})$, $n_i \geq 4$, and $V = n_1 n_2 n_3$. Cochain norms use counting measure. The photon configuration space is

$$\mathrm{Conf}(C_{\mathbf{n}}) = \{(A, z) : A \in C^1(C_{\mathbf{n}}; \mathbb{R}),\ z \in C^2(C_{\mathbf{n}}; \mathbb{Z}),\ dz = 0\}/\sim,$$

where $(A, z) \sim (A + r_q k, z - dk)$ for integer one-cochains k. Its curvature and phases are

$$F = dA + r_q z, \qquad u_b = e^{iqA(b)}. \tag{36}$$

Every closed integer flux is included. Quotient integration means

$$\int_{\mathrm{Conf}} \Phi = \sum_{dz=0} \int_{[0,r_q)^{3V}} \Phi(A, z)\, dA. \tag{37}$$

Any measurable fundamental domain for the same integer shifts gives the same integral. For a real link cochain,

$$(dA)_{ij}(x) = A_i(x) + A_j(x + e_i) - A_i(x + e_j) - A_j(x).$$

Let O_{cub} be the group of signed coordinate permutations, $W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}$, and fix $v_0 = (v_1, v_2, v_3)$ with $0 < v_3 < v_2 < v_1 < 1/2$. The fermion sites are $\mathcal{X} = Wv_0$. For $x = gv_0$ define $x^{(i)} = gs_iv_0$, where

$$s_0x = (x_1, x_2, -x_3), \quad s_1x = (x_1, x_3, x_2),$$
$$s_2x = (x_2, x_1, x_3), \qquad s_3x = (1 - x_1, x_2, x_3).$$

Every site has a unique expression $x = k + av_0$, $k \in \mathbb{Z}^3$, $a \in O_{\mathrm{cub}}$. Put $\pi(x) = k$. Types $0, 1, 2$ join sites with the same projection; type 3 joins nearest-neighbor photon sites. There are 48 fermion sites per photon site and eight type-3 bonds over each photon link.

The coordinate geometry needed below can be read directly from these formulas. In a fixed coordinate, the offsets of the fermion sites are $-v_1, -v_2, -v_3, v_3, v_2, v_1$. A type-0 bond changes the sign of the smallest absolute coordinate. Types $1, 2$ exchange consecutive absolute coordinates, and type 3 changes the largest positive offset to the largest negative one in the next cell. Thus a bond crossing a coordinate plane at an integer or half-integer joins opposite points at equal distance from that plane and is normal to it. No site lies on such a plane. These facts permit the exact half-slice decomposition in the next section.

Use the Pauli matrices σ_μ and the Hermitian Clifford triple $\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu)$. For $v \in \mathbb{R}^3$ set $\gamma_v = \sum_\mu v_\mu\gamma_\mu$. Write $A(y, x) = -A(x, y)$ and $A(x, x) = 0$. The operator is

$$(\mathcal{D}_A\psi)(x) = M\psi(x) - \sum_{i=0}^{3} \frac{w_i}{2} \left(\rho - \gamma_{\frac{x^{(i)}-x}{|x^{(i)}-x|}}\right) e^{iqA(\pi(x),\pi(x^{(i)}))}\psi(x^{(i)}). \tag{38}$$

We impose antiperiodic spinors in all three coordinate periods. The finite matrix has size $192V$. Its scalar lattice mass is $M - \rho\sum_i w_i/2$; no sign condition is imposed on it. For an oriented bond (x, x') of type i, abbreviate its hopping matrix and phase by

$$w(x, x') = \frac{w_i}{2} \left(\rho - \gamma_{\frac{x'-x}{|x'-x|}}\right), \qquad U_A(x, x') = e^{iqA(\pi(x),\pi(x'))}.$$

Introduce Grassmann generators $\psi_\alpha(x), \bar\psi_\alpha(x)$ with the same antiperiodic conventions. Normalize the Berezin integral by

$$\int \prod_x \prod_{\alpha=1}^{4} \psi_\alpha(x)\bar\psi_\alpha(x) = 1.$$

The pairs have even degree, so their order is immaterial. If $S(A) = \sum_x \bar\psi(x)(\mathcal{D}_A\psi)(x)$, then $\int e^{-S(A)} = \det \mathcal{D}_A$. For a real torus two-cochain b set

$$Z_{\mathbf{n}}(b) = \int_{\mathrm{Conf}(C_{\mathbf{n}})} e^{-\|F+b\|^2/2} \det \mathcal{D}_A \, d(A, z). \tag{39}$$

B.2. **Positivity at every real mass.**

Proposition B.1. For parameters (35) and real A, b, $\det \mathcal{D}_A \geq 0$ and $0 < Z_{\mathbf{n}}(b) < \infty$. ∎

Proof. Define

$$H_A\psi(x) = \sum_i w_iU_A(x, x^{(i)})\psi(x^{(i)}), \qquad J_A\psi(x) = \sum_i w_i\gamma_{\frac{x^{(i)}-x}{|x^{(i)}-x|}} U_A(x, x^{(i)})\psi(x^{(i)}).$$

Reversing a bond conjugates its phase and reverses its unit direction, so $H_A^* = H_A$ and $J_A^* = -J_A$. In the two Pauli blocks, H_A has identical blocks and J_A has opposite blocks. Since $\mathcal{D}_A = M - \rho H_A/2 + J_A/2$, it follows that

$$\mathcal{D}_A = \mathrm{diag}(B_A, B_A^*), \qquad \det \mathcal{D}_A = |\det B_A|^2 \geq 0. \tag{40}$$

Each type of hopping is a unitary permutation with phase followed by a matrix of norm $w_i(1+\rho)/2$. Hence

$$\|\mathcal{D}_A\| \le K_M := |M| + (1+\rho)\sum_i w_i/2.$$

To prove strict normalization for every M, it remains to show that the determinant is not identically zero on the phase torus. Regard it as a Laurent polynomial in the independent phases u_b. Fix a coordinate direction μ. The coefficient of $\prod_{b\|e_\mu} u_b^{32}$ is

$$C_*^V, \qquad C_* = (w_0/2)^{32}(w_1/2)^{64}(w_2/2)^{64}(w_3/2)^{32}(1-\rho^2)^{96} > 0. \tag{41}$$

The six offset layers listed above each contain $8V$ sites. A site has one upward neighbor, one downward neighbor and two neighbors in the same layer. The upward map is a permutation; its bond types through the six layers are $(2,1,0,1,2,3)$. Only the final upward step has a nontrivial phase in the chosen direction.

There are eight lower ends over a photon link, each with four spin rows. Achieving exponent 32 at that link forces all these rows to choose the upward type-3 block and excludes every reverse choice. At all links together these choices occupy all columns in the first layer. A first-layer row can then use neither its own layer nor its forbidden downward neighbor, so it must go upward. Proceeding through the layers forces every row to take its upward block. No mass entry or other link phase is selected.

The site-permutation sign occurs to the fourth power and is therefore one. Summing the remaining spin permutations gives the product of the 4-by-4 hopping determinants, $(w_i/2)^4(1-\rho^2)^2$. The boundary signs also occur to the fourth power. There are $8V$ sites in each layer, with the types just listed. Their product is (41).

For a nonnegative Laurent polynomial P on a phase torus, every Fourier coefficient has modulus at most its mean, since $|\int P(u)u^{-k}dH(u)| \le \int P(u)dH(u)$. Consequently,

$$\int_{[0,r_q)^{3V}} \det \mathcal{D}_A \, dA \ge r_q^{3V} C_*^V > 0.$$

On this phase box, $|(dA)_p| \le 2r_q$. The term $z=0$ in the configuration integral therefore proves $Z_{\mathbf{n}}(0) \ge r_q^{3V} e^{-6Vr_q^2} C_*^V > 0$. For any real b, the shifted Gaussian remains strictly positive on the same positive-measure determinant set, so $Z_{\mathbf{n}}(b) > 0$.

For finiteness, the one-dimensional Gaussian sum satisfies

$$C_q := \sup_{y\in\mathbb{R}} \sum_{k\in\mathbb{Z}} e^{-(y+r_qk)^2/2} \le 2 + r_q^{-1}\sqrt{2\pi} < \infty.$$

Retain the two terms nearest zero and bound the remaining terms by integrals over adjacent intervals of length r_q closer to zero. Those intervals have disjoint interiors, giving the bound. For each fixed A, dropping $dz=0$ in the nonnegative flux sum then bounds every shifted plaquette factor independently by C_q. Thus

$$Z_{\mathbf{n}}(b) \le r_q^{3V} C_q^{3V} K_M^{192V} < \infty.$$

□

At $b=0$, division of the density in (39) by $Z_{\mathbf{n}}(0)$ defines the unshifted probability measure. Write $\mathbb{E}_{\mathbf{n}}$ for expectation under this measure. Its curvature law is supported on $dF=0$. The matrix may be singular at some phases; pointwise invertibility was not needed.

B.3. **Gauge invariance and finite symmetries.** For a torus gauge transformation $h(x) = e^{i\chi(x)}$, replace $A(x,y)$ by $A(x,y) + (\chi(x) - \chi(y))/q$. Its curvature is unchanged. Returning each link to its phase cell changes z by an integer exact cochain and preserves the quotient measure. Multiplication of a chamber spinor at x by $h(\pi(x))$ conjugates $\mathcal{D}_A$ to the transformed operator. Hence its determinant

and the joint photon law are gauge invariant. The multiplier is periodic and therefore preserves the antiperiodic spinor conditions.

The finite law is invariant under translations. It is also covariant under permutations of the coordinate directions, with the rectangle and source cochain permuted at the same time. To check the spin part, let $\gamma_5 = \left(\begin{smallmatrix} 0 & I_2 \\ I_2 & 0 \end{smallmatrix}\right)$. It anticommutes with every γ_μ. If $r_n v = v - 2(v \cdot n)n$ for a unit vector n, the Clifford relations give

$$(\gamma_n\gamma_5)\gamma_v(\gamma_n\gamma_5)^{-1} = \gamma_{r_n v}.$$

Products of these matrices implement any signed coordinate permutation on the Clifford triple. The corresponding permutation of the sites preserves the four bond types and the projection π. It therefore intertwines the operators (38). The antiperiodic character is carried to the antiperiodic character on the permuted rectangle.

The photon change of variables permutes cochain coordinates with their orientation signs, preserves closedness and the counting norm, and maps a fundamental domain of (37) to another one. Taking determinants gives, in particular, the exact covariance of $Z_{\mathbf{n}}(b)$ under exchange of directions one and two. This is the finite covariance used below.

For charge conjugation put $C = \sigma_2 \oplus \sigma_2$. The identity $C\gamma_v^\top C^{-1} = -\gamma_v$, combined with bond reversal, gives $C\mathcal{D}_A^\top C^{-1} = \mathcal{D}_{-A}$. Thus $(A, z) \mapsto (-A, -z)$ preserves the normalized weight and sends F to $-F$. On cubic tori these observations give translations, signed coordinate permutations, and charge conjugation.

Appendix C. Finite fermion matrices for a coordinate slice

We factor the fermion determinant before integrating the photon field. The resulting matrices act on finite-dimensional spaces. Their entries are bounded functions of the spatial link phases, and the temporal link variables enter through diagonal unitaries. These are the properties needed for the Gaussian estimate below.

Write $\Sigma = \mathbb{Z}/n_2\mathbb{Z} \times \mathbb{Z}/n_3\mathbb{Z}$, and let $\Sigma^{(1)}$ be its set of positively oriented spatial links, so $|\Sigma^{(1)}| = 2|\Sigma|$. Distinguish the spatial connection A_j at time j from the temporal links a_j from time j to time $j+1$. The chamber sites in one photon cell have the six possible first coordinates

$$j - v_1, \quad j - v_2, \quad j - v_3, \quad j + v_3, \quad j + v_2, \quad j + v_1, \qquad 0 < v_3 < v_2 < v_1 < \tfrac{1}{2}.$$

Thus the open half-slab $H_j = \{x : j < x_1 < j + \frac{1}{2}\}$ contains three of these layers. Every photon projection of a site of H_j has time coordinate j. The restriction $S_{H_j}(A_j)$ of the fermion action to mass terms and bonds contained in H_j consequently depends only on A_j.

For a coordinate cut $x_1 = c$, $c \in \frac{1}{2}\mathbb{Z}$, put $r_c x = (2c - x_1, x_2, x_3)$. We use the antilinear, product-reversing operation $\mathcal{R}_c$ defined by

$$\mathcal{R}_c\psi_\alpha(x) = -\sum_\beta \bar\psi_\beta(r_c x)(\gamma_1)_{\beta\alpha}, \qquad \mathcal{R}_c\bar\psi_\alpha(x) = -\sum_\beta (\gamma_1)_{\alpha\beta}\psi_\beta(r_c x). \tag{42}$$

On scalar coefficients it conjugates the coefficient and pulls back the connection by r_c. The Clifford relations give $\mathcal{R}_c^2 = 1$ and

$$\mathcal{R}_c(\bar\psi(x)Y\psi(x')) = \bar\psi(r_c x')\gamma_1 Y^*\gamma_1\psi(r_c x). \tag{43}$$

In particular, it carries the action in one half-slab to the action in its reversed half-slab: the mass is real, and the hopping matrices satisfy $\gamma_1 w(x, x')\gamma_1 = w(r_c x', r_c x)$. The Berezin convention gives, for a finite set of sites Y,

$$\int_{r_c Y} \mathcal{R}_c\xi = \overline{\int_Y \xi}. \tag{44}$$

It suffices to check the top-degree monomial. At one site its coefficient is multiplied by $(\det\gamma_1)^2 = 1$, and the top-degree monomials at different sites have even degree.

We next factor each bond crossing the cut. Let x be its lower endpoint, $x' = r_c x$ its upper endpoint, and t its chamber bond type. With $P^{\pm} = (1 \pm \gamma_1)/2$ the two hopping matrices are

$$\begin{aligned} w(x,x') &= \tfrac{w_t}{2}\big((1+\rho)P^- - (1-\rho)P^+\big), \\ w(x',x) &= \tfrac{w_t}{2}\big((1+\rho)P^+ - (1-\rho)P^-\big). \end{aligned} \tag{45}$$

Choose orthonormal bases $e_1^{\pm}, e_2^{\pm}$ of the ranges of $P^{\pm}$. For each $k = 1, 2$ use the following four modes at the upper endpoint; $U = U_A(x,x')$ denotes the bond phase.

	ζ	ω	u
1	$(e_k^-)^*\psi(x')$	$w_t(1+\rho)/2$	U
2	$(e_k^+)^*\psi(x')$	$w_t(1-\rho)/2$	U
3	$\bar\psi(x')e_k^+$	$w_t(1+\rho)/2$	$\overline{U}$
4	$\bar\psi(x')e_k^-$	$w_t(1-\rho)/2$	$\overline{U}$

(46)

Their reversed modes, in the same order, are

$$\bar\psi(x)e_k^-, \qquad -\bar\psi(x)e_k^+, \qquad -(e_k^+)^*\psi(x), \qquad (e_k^-)^*\psi(x).$$

Multiplication by the original mode and anticommutation reproduce the four projected terms in (45). Consequently, if $S_c^{\times}$ is the part of the action on all bonds crossing the cut, then

$$-S_c^{\times} = \sum_{i \in \mathcal{J}(c)} \omega_i u_i (\mathcal{R}_c \zeta_i)\zeta_i. \tag{47}$$

Here $\mathcal{J}(c)$ is the finite set of crossing modes, with the transverse antiperiodic identifications already imposed. Fix an order on this set, and for $I \subset \mathcal{J}(c)$ write $\zeta_I = \prod_{i\in I}\zeta_i$, $\omega_I = \prod_{i\in I}\omega_i$, and $u_I = \prod_{i\in I} u_i$. Every summand on the right of (47) is even and has square zero. Distinct summands commute; reversal of order under $\mathcal{R}_c$ cancels the sign incurred in collecting the original modes. Hence

$$e^{-S_c^{\times}} = \sum_{I \subset \mathcal{J}(c)} \omega_I u_I \mathcal{R}_c(\zeta_I)\zeta_I. \tag{48}$$

All ω_i are strictly positive because $w_t > 0$ and $0 < \rho < 1$. At a site cut $c = j$ the phase is one. At the link cut $c = j + \frac{1}{2}$, groups one and two in (46) have charge $+1$ and groups three and four have charge -1. If $\mathfrak{q}(I) \in \mathbb{Z}^{\Sigma}$ is the sum of these charges over each temporal photon link, then $u_I = \exp(iq\langle a_j, \mathfrak{q}(I)\rangle)$.

All slab endpoints below belong to $\frac{1}{2}\mathbb{Z}$. Let $\mathcal{F}(c)$ be the exterior space on the crossing-mode labels, with orthonormal subset basis $(e_I)_{I\subset\mathcal{J}(c)}$. For an even Grassmann element B supported in $c < x_1 < c'$, define the matrix

$$N_{c,c'}[B]_{I',I} = \int_{c<x_1<c'} \zeta_I^{(c)} B \mathcal{R}_{c'}(\zeta_{I'}^{(c')}). \tag{49}$$

A parity-preserving matrix T on $\mathcal{F}(c)$ is inserted at the cut by

$$\mathrm{Ins}_c[T] = \sum_{I,J} T_{IJ}\mathcal{R}_c(\zeta_J^{(c)})\zeta_I^{(c)}.$$

These conventions make composition ordinary matrix multiplication:

$$N_{c,c''}[B\,\mathrm{Ins}_{c'}[T]B'] = N_{c',c''}[B']T N_{c,c'}[B]. \tag{50}$$

Indeed, an entry on the left is a sum of products

$$[\zeta_I^{(c)} B \mathcal{R}_{c'}\zeta_J^{(c')}][\zeta_K^{(c')} B' \mathcal{R}_{c''}\zeta_{I''}^{(c'')}]T_{KJ}.$$

The bracketed factors involve disjoint slab generators. Their Berezin integrals factor without permuting the displayed factors, giving the right side of (50). Equations (43) and (44) also give

$$N_{c',2c'-c}[\mathcal{R}_{c'}B] = N_{c,c'}[B]^*, \tag{51}$$

after identifying the outer crossing-mode spaces by normal translation. For the half-slabs used here this translation is by one lattice unit; choose the mode orders by this translation, so that it introduces no reordering sign. Reversing the integrand reverses its two boundary factors and conjugates its Berezin coefficient, which is precisely the adjoint on the right.

Denote the site and link crossing-mode spaces by $\mathcal{F}_s$ and $\mathcal{F}_l$. The half-slab matrix is

$$N(A_j) = N_{j,j+1/2}[e^{-S_{H_j}(A_j)}] : \mathcal{F}_s \longrightarrow \mathcal{F}_l.$$

Let $\Omega_s e_I = \omega_I e_I$ and $\Omega_l e_J = \omega_J e_J$ be the two positive diagonal crossing weights, and put

$$\widehat{N}(A) = \Omega_l^{1/2} N(A) \Omega_s^{1/2}, \qquad u(a)e_I = e^{iq\langle a, \mathfrak{q}(I)\rangle} e_I \quad \text{on } \mathcal{F}_l. \tag{52}$$

The matrices $N(A)$ preserve parity: a nonzero entry in (49) requires $|I| + |I'|$ to be even, since the action and the top-degree Berezin monomial have even degree. The matrix $u(a)$ is unitary. Every entry of $N(A)$ is a polynomial in the spatial link phases and their conjugates: the exponential of the quadratic Grassmann action terminates, and integration extracts one coefficient. The finite cross-section therefore gives

$$\sup_{A \in \mathbb{R}^{\Sigma^{(1)}}} \|\widehat{N}(A)\| < \infty, \qquad \widehat{N}(A + r_q k) = \widehat{N}(A), \quad r_q = 2\pi/q, \quad k \in \mathbb{Z}^{\Sigma^{(1)}}. \tag{53}$$

No uniformity in the cross-section is asserted in this bound.

Proposition C.1. For the finite rectangular torus, with antiperiodic fermions in the time direction,

$$\det \mathcal{D}_A^{C_{\mathbf{n}}} = \operatorname{tr}_{\mathcal{F}_s} \prod_{j=n_1-1,\dots,0} \big(\widehat{N}(A_{j+1})^* u(a_j) \widehat{N}(A_j)\big). \tag{54}$$

Here $A_{n_1} = A_0$. The same identity holds with $A_{n_1} = A_0 + g$ for any $g \in r_q \mathbb{Z}^{\Sigma^{(1)}}$. ■

Proof. Split the fermion action at every site cut and link cut. The half-slab below $j + \frac{1}{2}$ has matrix $N(A_j)$, while the one above it has matrix $N(A_{j+1})^*$ by (51). The crossing expansions (48) insert Ω_s at a site cut and $\Omega_l u(a_j)$ at a link cut. Composition therefore gives, before taking the cyclic trace, the successive factors

$$N(A_{j+1})^* \Omega_l u(a_j) N(A_j) \Omega_s.$$

These half-slab and crossing actions contain each mass and hopping term exactly once.

We verify the closing sign. Fix the crossing subset I at the initial site cut. Translating its product to the final cut multiplies each generator by the time character $\chi = -1$. Moving this product across the adjacent boundary product contributes $(-1)^{|I|}$ in each term of even total degree. Thus the closing factor is $(-\chi)^{|I|} = 1$. The normal-translation ordering of the modes introduces no further sign. Closing the Berezin integral therefore takes the ordinary matrix trace. Its value is $\int e^{-S(A)} = \det \mathcal{D}_A^{C_{\mathbf{n}}}$.

The diagonal matrices Ω_l and $u(a_j)$ commute. Distribute $\Omega_l^{1/2}$ to the two adjoining half-slab matrices and use cyclicity of the trace to distribute $\Omega_s^{1/2}$ at every site cut. This gives (54). Finally, (53) identifies $\widehat{N}(A_0 + g)$ with $\widehat{N}(A_0)$, proving the assertion for the stated shifted endpoint. □

Appendix D. The all-flux trace and Gaussian domination

D.1. **Unfolding the photon variables.** Take coordinate one as time and set $\Sigma = (\mathbb{Z}/n_2\mathbb{Z}) \times (\mathbb{Z}/n_3\mathbb{Z})$, $v = |\Sigma|$. Write d_s for its spatial coboundary, A_j for its $2v$ spatial link variables at time j, and a_j for the v temporal link variables from j to $j+1$. In phase-cell coordinates the electric and magnetic curvatures are

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_s a_j + r_q z_j^{\mathrm{el}}, \qquad F_j^{\mathrm{mag}} = d_s A_j + r_q z_j^{\mathrm{mag}}. \tag{55}$$

The electric components are the 12 and 13 components; the magnetic component is 23. Closedness of z reads $z_{j+1}^{\mathrm{mag}} - z_j^{\mathrm{mag}} = d_s z_j^{\mathrm{el}}$.

Set $k_0 = 0$ and $k_{j+1} = k_j + z_j^{\mathrm{el}}$. Then

$$z_j^{\mathrm{mag}} = z_0^{\mathrm{mag}} + d_s k_j, \qquad d_s k_{n_1} = 0.$$

Replace A_j by $A_j + r_q k_j$. Summation over the independent integers k_j, $0 < j < n_1$, unfolds their phase boxes to $\mathbb{R}^{\Sigma^{(1)}}$. The last integer gives a winding

$$g = r_q k_{n_1} \in \Gamma_s, \qquad \Gamma_s = r_q\big(\mathbb{Z}^{\Sigma^{(1)}} \cap \ker d_s\big).$$

It is retained as the endpoint condition $A_{n_1} = A_0 + g$.

Fix a spatial plaquette σ_0, and let $k = \sum_\sigma z_0^{\mathrm{mag}}(\sigma)$. There is an integer one-cochain k' with $z_0^{\mathrm{mag}} - k\mathbf{1}_{\sigma_0} = d_s k'$. Use the connected dual graph of the spatial torus, whose vertices are plaquettes. A signed unit flow along a path from σ_0 to σ has divergence $\mathbf{1}_\sigma - \mathbf{1}_{\sigma_0}$. A spanning tree supplies such paths for all σ. Adding them with the coefficients of any zero-sum integer two-cochain realizes that cochain as the divergence of an integer flow, which is its spatial coboundary. Two choices of k' differ by an integer element of $\ker d_s$.

Shift every spatial slice by $r_q k'$. Summing k' modulo closed integer one-cochains unfolds the initial phase box to a fundamental domain $D_s \subset \mathbb{R}^{\Sigma^{(1)}}$ of Γ_s. Put $Q_s = \mathbb{R}^{\Sigma^{(1)}}/\Gamma_s$, with its induced Lebesgue measure. In the path integral we represent its initial point by the actual vector $A_0 \in D_s$; the endpoint $A_0 + g$ is taken in the covering vector space. Thus the configuration integral becomes a sum over $k \in \mathbb{Z}$, $g \in \Gamma_s$, and an integral over

$$A_0 \in D_s, \quad A_j \in \mathbb{R}^{\Sigma^{(1)}}\ (0 < j < n_1), \quad a_j \in [0, r_q)^\Sigma, \quad A_{n_1} = A_0 + g. \tag{56}$$

The curvatures in these variables are

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_s a_j, \qquad F_j^{\mathrm{mag}} = d_s A_j + r_q k \mathbf{1}_{\sigma_0}. \tag{57}$$

All measures are induced by Lebesgue measure, with counting measure on the two discrete sums. No magnetic flux or electric winding has been removed.

These transformations are bijective reindexings of the integer sums followed by translations of phase boxes. They preserve both link phases and real curvatures. Tonelli's theorem proves the integral identity for nonnegative functions. Applying it first to the absolute value proves the same identity for absolutely integrable complex functions. It therefore applies to the partition integrand and to the determinant trace (54).

D.2. **Slice operators and their integrability.** Use the finite spaces $\mathcal{F}_s, \mathcal{F}_l$ and $\widehat{N}(A)$ constructed in the preceding section. On the unfolded photon space put

$$H_s = \bigoplus_{k\in\mathbb{Z}} L^2(Q_s; \mathcal{F}_s), \qquad H_l = \bigoplus_{k\in\mathbb{Z}} L^2(Q_s; \mathcal{F}_l).$$

The translation generators $E_b = -i\partial_{A_b}$ are specified by $e^{-i\langle x, E\rangle} f(A) = f(A - x)$. On H_l define

$$\begin{aligned} (U_G(a)f)(A) &= u(a) f(A - d_s a), \\ P_G &= r_q^{-v} \int_{[0,r_q)^\Sigma} U_G(a)\, da, \qquad K = r_q^v e^{-|E|^2/2} P_G. \end{aligned} \tag{58}$$

The integer charges of $u(a)$ and the quotient by Γ_s make U_G a unitary representation of the compact group $\mathbb{R}^\Sigma/r_q\mathbb{Z}^\Sigma$. Its average P_G is an orthogonal projection: the group law gives $P_G^2 = P_G$, and invariance under $a \mapsto -a$ gives $P_G^* = P_G$. It commutes with all translations of A. Hence $K \geq 0$ and

$$K^{1/2} = r_q^{v/2} e^{-|E|^2/4} P_G. \tag{59}$$

For a real magnetic slice source b define

$$(M_b f)_k(A) = e^{-\|d_s A + r_q k \mathbf{1}_{\sigma_0} + b\|^2/4} \widehat{N}(A) f_k(A), \qquad X_b = K^{1/2} M_b. \tag{60}$$

Here $M_b : H_s \to H_l$ is bounded. The operator X_b will be Hilbert–Schmidt even though Q_s is noncompact and the flux index k is unbounded.

Lemma D.1. For every real spatial two-cochain b, $X_b \in \mathfrak{S}_2(H_s, H_l)$. The constants in this assertion may depend on the finite cross-section and on r_q. ■

Proof. The heat kernel of $e^{-|E|^2/4}$ is the periodization

$$g([x]) = \sum_{\gamma \in \Gamma_s} \pi^{-v} e^{-|x+\gamma|^2}.$$

Its integral on Q_s is one. Since $\Gamma_s \subset r_q \mathbb{Z}^{2v}$, its supremum is bounded by the product of $2v$ one-dimensional Gaussian sums. Thus $g \in L^1(Q_s) \cap L^\infty(Q_s)$. The matrix kernel of (59) is

$$K^{1/2}(A', A) = r_q^{-v/2} \int_{[0,r_q)^\Sigma} g([A' - A - d_s a]) u(a)\, da.$$

Its norm is bounded by $\widehat{g}([A' - A])$, where $\widehat{g}([x]) = r_q^{-v/2} \int g([x - d_s a])\, da$. Compact averaging preserves finite L^1 and L^∞ norms, so $\widehat{g} \in L^2$.

Let $C_N = \sup_A \|\widehat{N}(A)\|$. Integrating the squared kernel in A' gives

$$\|X_b\|_2^2 \le (\dim \mathcal{F}_s) C_N^2 \|\widehat{g}\|_2^2 \sum_{k \in \mathbb{Z}} \int_{Q_s} e^{-\|d_s A + r_q k \mathbf{1}_{\sigma_0} + b\|^2/2}\, dA. \tag{61}$$

The remaining integral separates into $\ker d_s/\Gamma_s$ and $(\ker d_s)^\perp$. The first factor is compact. Indeed, d_s is an integer matrix; Gaussian elimination over the rationals gives a rational basis of its real kernel, and clearing denominators gives a full-rank integer lattice in that kernel. On the second factor d_s is injective.

The adjoint equations on the two-dimensional periodic slice are, up to the fixed orientation convention,

$$(d_s^* b)(y, 2) = b(y) - b(y - e_3), \qquad (d_s^* b)(y, 3) = b(y - e_2) - b(y).$$

Thus $\ker d_s^*$ consists exactly of constant plaquette functions. The component of $r_q k \mathbf{1}_{\sigma_0} + b$ perpendicular to $\operatorname{ran} d_s$ has squared norm $v^{-1}(r_q k + \sum_\sigma b_\sigma)^2$. A translation in $(\ker d_s)^\perp$ removes the component in $\operatorname{ran} d_s$. It follows that

$$\sum_k \int_{Q_s} e^{-\|d_s A + r_q k \mathbf{1}_{\sigma_0} + b\|^2/2}\, dA = C_s \sum_{k \in \mathbb{Z}} e^{-(r_q k + \sum_\sigma b_\sigma)^2/(2v)} < \infty, \tag{62}$$

where

$$C_s = \operatorname{vol}(\ker d_s/\Gamma_s) \int_{(\ker d_s)^\perp} e^{-\|d_s A\|^2/2}\, dA < \infty.$$

The finite-dimensional quadratic form in this last integral is positive definite. Equations (61) and (62) prove the lemma. □

D.3. **An identity of absolutely convergent traces.** For a real source b on $C_{\mathbf{n}}$ write b_j^{el} and b_j^{mag} for its slice components, and put $X_j = X_{b_j^{\mathrm{mag}}}$, $U_j = e^{i\langle b_j^{\mathrm{el}}, E\rangle}$. Each U_j is unitary and commutes with $K^{1/2}$.

Proposition D.2. The shifted partition function is

$$Z_{\mathbf{n}}(b) = (2\pi)^V \operatorname{Tr}_{H_s} \prod_{j = n_1 - 1, \dots, 0} (X_{j+1}^* U_j X_j), \qquad X_{n_1} = X_0. \tag{63}$$

The product is trace class. The corresponding cyclic kernel integral and all its periodization and flux sums converge absolutely. The construction holds with any coordinate chosen as time. ■

Proof. We first identify the kernels, then prove absolute convergence of their cyclic integral. For a real electric slice source b, the kernel of $e^{i\langle b,E\rangle}K$ is

$$\mathscr{K}_b(A',A) = (2\pi)^{-v}\int_{[0,r_q)^\Sigma}\sum_{\gamma\in\Gamma_s} e^{-|A'-A-d_sa+b+\gamma|^2/2}u(a)\,da. \tag{64}$$

Indeed, $(2\pi)^{-v}e^{-|x|^2/2}$ is the normalized heat kernel in $2v$ coordinates, and the temporal integral is the factor $r_q^v P_G$ in K. Our translation convention gives $e^{i\langle b,E\rangle}f(A') = f(A'+b)$, fixing the sign of b. The kernel acts separately in each magnetic flux sector.

In (56) write $A_j = A_j^\flat + g_j$, with $A_j^\flat$ in a fixed fundamental domain of Γ_s, $g_0 = 0$, and $g_{n_1} = g$. Then the electric differences contain $g_{j+1} - g_j$. The map

$$(g_1,\dots,g_{n_1}) \longmapsto (g_{j+1}-g_j)_{0\le j<n_1}$$

is a bijection on $\Gamma_s^{n_1}$. Hence summing these differences independently is precisely periodization of each cyclic electric kernel. The magnetic terms and $\widehat{N}$ are unchanged by these shifts.

For a fixed flux k, write

$$M_{j,k}(A) = e^{-\|d_sA+r_qk\mathbf{1}_{\sigma_0}+b_j^{\mathrm{mag}}\|^2/4}\widehat{N}(A).$$

Splitting each magnetic Gaussian into two equal factors and using (54) gives the cyclic integral

$$(2\pi)^V\sum_{k\in\mathbb{Z}}\int_{Q_s^{n_1}}\mathrm{tr}_{\mathcal{F}_s}\prod_{j=n_1-1,\dots,0}\Big[M_{j+1,k}(A_{j+1})^*\mathscr{K}_{b_j^{\mathrm{el}}}(A_{j+1},A_j)M_{j,k}(A_j)\Big]\prod_{j=0}^{n_1-1}dA_j, \tag{65}$$

where $A_{n_1} = A_0$ and $M_{n_1,k} = M_{0,k}$. Each slice contributes the factor $(2\pi)^v$, so the prefactor is $(2\pi)^{vn_1} = (2\pi)^V$. Since U_j commutes with $K^{1/2}$,

$$M_{j+1}^*U_jKM_j = X_{j+1}^*U_jX_j.$$

Thus (65), once justified, is exactly the trace in (63).

For an absolute majorant replace finite matrix entries by their norms. Use $|x+y|^2 \ge |x|^2/2 - |y|^2$ to absorb any fixed source into a Gaussian with half its exponent. The temporal variables range over a compact box. Their shifts can be absorbed in the same way, with a finite constant depending on the fixed cross-section and source. The resulting scalar electric kernels are nonnegative periodized Gaussians with finite L^2 norm; the scalar magnetic factors are square-integrable after summation over k, by the proof of (62) with the altered positive Gaussian coefficient.

Thus the scalar operators obtained by alternating one electric kernel and one magnetic factor are Hilbert–Schmidt. To bound their cyclic integral, compose all but one of the kernels. The resulting operator is Hilbert–Schmidt, since a Hilbert–Schmidt operator is bounded and multiplication by bounded operators preserves the Hilbert–Schmidt class. Cauchy–Schwarz for the last two kernels bounds the cyclic integral by the product of their Hilbert–Schmidt norms. For the nonnegative kernels all compositions and sums are justified by Tonelli. Fubini then applies to the original matrix integrand.

Lemma D.1 also shows directly that $X_{j+1}^*U_jX_j$ is trace class. The absolute majorant just proved identifies its cyclic trace with the original path integral and permits cyclicity, including between the two Hilbert spaces. □

D.4. **Removing the external source.** The Hilbert–Schmidt bound justifies the trace formula. The following comparison removes the source without using the size of that bound.

Proposition D.3. For every rectangle with $n_i \ge 4$ and every real source b,

$$0 < Z_{\mathbf{n}}(b) \le Z_{\mathbf{n}}(0). \tag{66}$$

Consequently, the unshifted curvature law satisfies

$$\mathbb{E}_{\mathbf{n}}e^{\langle b,F\rangle} \le e^{\|b\|^2/2}. \tag{67}$$

The constant is independent of the volume and the parameters (35). ■

Proof. Strict positivity was proved in proposition B.1. Move the final boundary map in (63) through the trace to write

$$(2\pi)^{-V} Z_{\mathbf{n}}(b) = \operatorname{Tr}_{H_l}(U_{n_1-1} X_{n_1-1} X_{n_1-1}^* \cdots U_0 X_0 X_0^*).$$

For operators $Y_1, \dots, Y_m$, Schatten Hölder gives $|\operatorname{Tr}(Y_1 \cdots Y_m)| \le \prod_j \|Y_j\|_m$ when the displayed norms are finite; see [Pon25, slides 53–56]. Here $X_j X_j^*$ is trace class, hence belongs to $\mathfrak{S}_{n_1}$, and multiplication by U_j does not change its Schatten norm. The nonzero spectra of $X_j X_j^*$ and $X_j^* X_j$ agree. Therefore

$$Z_{\mathbf{n}}(b) \le \prod_{j=0}^{n_1-1} \left((2\pi)^V \operatorname{Tr}((X_j^* X_j)^{n_1}) \right)^{1/n_1} = \prod_{j=0}^{n_1-1} Z_{\mathbf{n}}(b^{[j]})^{1/n_1}, \tag{68}$$

where $b^{[j]}$ has zero electric part and repeats the magnetic cochain b_j^{mag} in every slice. The equality follows by applying the exact trace formula to this repeated source; its unitary factors are all the identity.

Every $b^{[j]}$ is supported only on 23-plaquettes. Exchange coordinates one and two, carrying the rectangle to $C_{(n_2,n_1,n_3)}$. The transported source is now purely electric for the new time coordinate. Its magnetic part is zero in every slice. Applying (68) in that direction bounds its partition function by the unperturbed one. The exact finite covariance proved above identifies these with $Z_{\mathbf{n}}(b^{[j]})$ and $Z_{\mathbf{n}}(0)$. Substitution into (68) proves (66).

Finally, completing the square in the real curvature gives

$$e^{\langle b,F\rangle} e^{-\|F\|^2/2} = e^{\|b\|^2/2} e^{-\|F-b\|^2/2}.$$

After integrating the positive determinant weight and dividing by $Z_{\mathbf{n}}(0)$, (66) with source $-b$ gives (67). □

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